

ES3J1 Advanced Systems and Software Engineering

Python Assignment for 2023–2024

T. J. Sullivan*

*** Revised Version of 23 February 2024 ***

A seaborne drone of mass $m = 1.5$ kg is adrift on a planar ocean; its position and momentum are $\mathbf{q} \in \mathbb{R}^2$ and $\mathbf{p} \in \mathbb{R}^2$ respectively. The drone is caught in the influence of a whirlpool centred at the origin.

The whirlpool is such that, when the drone is at position $\mathbf{q}$, the ocean flow imparts a force $\mathbf{f} \in \mathbb{R}^2$ on the drone proportional to the difference between its velocity $\mathbf{v}$ and the background flow velocity $\mathbf{v}_b = \beta R_{3\pi/4} \mathbf{q}$, where $\beta = 2 \text{ s}^{-1}$ and $R_\theta \in \mathbb{R}^{2 \times 2}$ denotes rotation through θ in the positive (anticlockwise) direction. The constant of proportionality between the force and the velocity difference, i.e. the drag coefficient, is $\gamma = 2.5 \text{ N} \cdot \text{s} \cdot \text{m}^{-1}$. That is,

$$\mathbf{f} = \gamma (\mathbf{v}_b - \mathbf{v}). \quad (\dagger)$$

It is also possible for the drone to exert a control force $\mathbf{u} \in \mathbb{R}^2$.

(Q1) Use Newton's laws of motion to derive a second-order ODE for the evolution of the position $\mathbf{q}$. Reformulate this second-order ODE for $\mathbf{q}$ into a linear first-order ODE (state-space model) for the state $\mathbf{x} = (\mathbf{q}, \mathbf{p}) \in \mathbb{R}^4$ of the form

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \quad (*)$$

for appropriate matrices $A \in \mathbb{R}^{4 \times 4}$ and $B \in \mathbb{R}^{4 \times 2}$.

Hint: You may find it helpful, both for these calculations and for your Python code, to write these matrices A and B in terms of 2×2 blocks corresponding to the position and momentum components of the state. Start by rewriting $(\dagger)$ in terms of $\mathbf{q}$ and $\mathbf{p}$. Some of the blocks of A and B are given to you below:

$$\begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{bmatrix} = \begin{bmatrix} 0_{2 \times 2} & m^{-1} I_{2 \times 2} \\ ? & ? \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \mathbf{p} \end{bmatrix} + \begin{bmatrix} ? \\ ? \end{bmatrix} \mathbf{u}. \quad (*)$$

(Q2) Implement the implicit Euler scheme for the numerical solution of the ODE $(*)$ with a constant time step, as in Lecture/Lab 4. Use this scheme in the special case $\mathbf{u} \equiv \mathbf{0}$, with a time step of 0.001 s , to approximate the motion of the drone, starting from rest at $\mathbf{q}_0 = (5, 5) \text{ m}$ at time $t_0 = 0$, up to time $t_{\max} = 30 \text{ s}$. Plot the evolution of the components of the state. Verify that the drone follows a path spiralling inwards towards the origin and that it drifts to within 2 m of the origin within around 15 s .

*Mathematics Institute and School of Engineering, University of Warwick, Coventry, CV4 7AL, United Kingdom (t.j.sullivan@warwick.ac.uk)

(Q3) At each integer time, a noisy observation $\mathbf{y} \in \mathbb{R}^2$ of the position $\mathbf{q}$ is made, with an additive Gaussian observational error of mean zero and standard deviation 0.5 m in each direction independently. Implement the linear Kalman filter to estimate the states $\mathbf{x}(t)$. (Note that the filter will need to take account of the control force $\mathbf{u}$, which is still zero for this part of the assignment but will be non-zero in the next part.) Initialise your filter with the incorrect state mean vector $\widehat{\mathbf{m}} = \mathbf{0}$ and highly agnostic state covariance matrix $\widehat{\mathbf{C}} = 10^6 I_{4 \times 4}$. Plot on appropriate axes the true states $\mathbf{x}$, the observed data $\mathbf{y}$, your filtered state estimates $\widehat{\mathbf{m}}$, and the predictive errors $\widehat{\mathbf{m}} - \mathbf{x}$; plot also a ± 1 standard deviation confidence region for each component of the state and comment on how this compares to the predictive error.

Hint: At least if one ignores the fact that $\mathbf{u}$ might depend on $\mathbf{x}$, the implicit Euler discretisation of (*) takes the form $\mathbf{x}(t_{i+1}) = F_i \mathbf{x}_{t_i}$ for a suitable matrix $F_i \in \mathbb{R}^{4 \times 4}$, as in Lecture/Lab 3. What is the observation matrix $H \in \mathbb{R}^{2 \times 4}$?

(Q4) The drone has a motor capable of exerting a force $\mathbf{u} \in \mathbb{R}^2$ of maximum magnitude $\|\mathbf{u}\| \leq 5$ N. Use your filtered state estimates and a linear feedback rule to bring the drone to rest at $\mathbf{q}_* = (-1, 1)$ m.

Notes: Ideally, your feedback rule would make the net force $\mathbf{f} + \mathbf{u}$ a linear function of $\mathbf{x}$ — and hence a “PD” control rule for $\mathbf{q}$. However, you will have to depart from this ideal in several ways:

- (i) The applied control force $\mathbf{u}$ will have to respect the maximum magnitude constraint.
- (ii) The feedback rule cannot rely on knowledge of the exact state $\mathbf{x}$, only the approximation $\widehat{\mathbf{m}} \approx \mathbf{x}$.
- (iii) Because the state estimate will not be perfectly accurate, it is unrealistic to bring the drone exactly to perfect rest at the desired position — you will have to set appropriate error tolerances for these goals.

The above four tasks are to be completed in a single Jupyter notebook. Submit the `.ipynb` with all cells (both Python code and Markdown) executed, along with a `.pdf` printout of the complete notebook. The two files are to be named with your University ID, e.g. `u1234567.ipynb` and `u1234567.pdf`, and submitted via Tabula. In order to ensure reproducibility of your results, the random number generator must be seeded with your University ID number.

Each of the above four tasks Q1–Q4 carries 20% of the available credit. The remaining 20% credit will be awarded for the quality of scientific communication (readability and commenting of code, clear visualisation of numerical data, etc.) and the general quality of the Python code (meeting or exceeding the best practices outlined in the taught component of the module). Conversely, major flaws such as non-functional Python code will attract a significant penalty.